Funica Project Guide: Konsi File Kya Kaam Karti Hai

Ye doc `create_funica_mvvm.ps1` se bane project ke har folder aur file ka kaam batata hai. Jab bhi confusion ho ki koi code kahan likhna hai, yahan dekh lo.

Quick Rule: Code Kahan Likhun?

Agar tumhe ye likhna hai	Ye file/folder
Screen ka design (button, text, layout)	`features/<feature>/views/`
Chhota reusable widget sirf ek feature ka	`features/<feature>/views/widgets/`
Widget jo kai features mein use hoga	`shared/widgets/`
Button click pe kya hoga, loading, error	`features/<feature>/viewmodels/`
Firebase/API se data lana ya bhejna	`features/<feature>/repositories/`
Data ka structure (Product, User)	`shared/models/` ya `features/<feature>/models/`
Colors, font, theme	`core/theme/`
Screens ke beech navigation	`core/router/app_router.dart`
Text ya image path constants	`core/constants/`
Helper functions (validation, format)	`core/utils/`

MVVM Flow (Ek Baar Samajh Lo)

```
View (screen)  ->  ViewModel  ->  Repository  ->  Firebase
```

- **View** user ko dikhata hai aur click events ViewModel ko bhejta hai. Isme business logic nahi hoti.
- **ViewModel** state rakhta hai (loading, error, data) aur logic chalata hai.
- **Repository** sirf data lata/bhejta hai. Firebase ka code sirf yahin hota hai.

Example: Login button dabaya, `LoginScreen` ne `AuthViewModel` ko call kiya, usne `AuthRepository` se Firebase login karwaya, result aaya, ViewModel ne state badli, aur screen apne aap update ho gayi.

Root Level (Project ke Bahar wali Files)

File / Folder	Kaam
`pubspec.yaml`	Project ka config: packages (dependencies), assets ke paths, app version. Naya package ya image folder yahin register hota hai.
`pubspec.lock`	Packages ke exact versions lock karta hai. Haath se mat badlo.
`analysis_options.yaml`	Dart lint rules (code style warnings).
`.gitignore`	Git ko batata hai kaunsi files push nahi karni (build folder, keys).
`README.md`	Project ka intro aur setup steps.
`lib/`	Saara Dart code yahin hai. 90% kaam yahin hoga.
`assets/`	Images, icons, illustrations. Inhe `pubspec.yaml` mein register karna padta hai.
`android/`	Android native project. Yahan `google-services.json` aur keystore config jaati hai.
`ios/`	iOS native project. Yahan `GoogleService-Info.plist` jaati hai.
`test/`	Unit aur widget tests.
`build/`	Auto-generated build output. Delete kar sakte ho, git mein nahi jaata.
`.dart_tool/`	Flutter ki internal cache. Chhedna mat.

`lib/` ke Andar

`lib/main.dart`

App ka entry point. Yahan `Firebase.initializeApp()`, `ProviderScope` (Riverpod ke liye), aur `MaterialApp` (theme aur home screen) set hota hai. Router lagane ke baad `MaterialApp.router` use hoga.

`lib/firebase_options.dart`

Ye `flutterfire configure` chalane ke baad apne aap banti hai. Isme Firebase project ke keys hote hain. Haath se edit mat karo.

`lib/core/` : Poore App ki Common Cheezein

Isme wo cheezein hain jo kisi ek feature ki nahi, poori app ki hain.

File / Folder	Kaam
`core/theme/app_colors.dart`	Saare colors ek jagah (background, surface, error, success). Color badalna ho to sirf yahan.
`core/theme/app_theme.dart`	ThemeData: dark mode, font (Urbanist), button style.
`core/router/app_router.dart`	go_router config: saare routes (login, home, cart...) aur bottom nav ka ShellRoute.
`core/constants/app_strings.dart`	App mein use hone wale fixed texts.
`core/constants/app_assets.dart`	Image aur icon ke paths ke constants, taaki typo na ho.
`core/utils/validators.dart`	Email, password, phone validation functions.
`core/utils/formatters.dart`	Price, date jaise formatting helpers.
`core/network/api_client.dart`	`dio` ka setup, agar kabhi REST API call karni ho.
`core/services/storage_service.dart`	Local storage (PIN, tokens, settings) ka wrapper.

`lib/shared/` : Kai Features Mein Use Hone Wali Cheezein

`shared/widgets/`

File	Kaam
`primary_button.dart`	White pill-shape button jo har screen pe use hota hai (Continue, Sign in, Add to Cart).
`app_text_field.dart`	Rounded input field (email, password, search).
`product_card.dart`	Product ka card: image, naam, rating, price, heart icon. Home, wishlist, search sab jagah.
`category_chip.dart`	Filter chips (All, Sofa, Chair...).
`success_dialog.dart`	Congratulations/success modal (signup, order, top up).

`shared/models/`

File	Kaam
`product.dart`	Product class: id, name, price, images, colors, rating. Firestore document se convert karne ke `fromMap` / `toMap` yahin.
`user.dart`	User profile ka structure.
`order.dart`	Order ka structure: items, total, status.
`address.dart`	Shipping address ka structure.

`lib/features/` : Asli App

Har feature ek folder hai aur uske andar chaar cheezein:

Subfolder	Kaam
`views/`	Us feature ki screens

Subfolder	Kaam
`views/widgets/`	Sirf us feature ke chhote widgets
`viewmodels/`	`<feature>_viewmodel.dart`: state class + Notifier + provider
`models/`	Us feature ke khaas models (jo shared mein nahi jaate)
`repositories/`	`<feature>_repository.dart`: abstract class, implementation, provider

Neeche har feature ki screens ka kaam hai.

`onboarding/`

Screen	Kaam
`splash_screen.dart`	App khulte hi logo dikhata hai, phir check karta hai ki user logged in hai ya nahi.
`onboarding_screen.dart`	Pehli baar wale user ke liye 3 intro slides.

`auth/`

Screen	Kaam
`lets_you_in_screen.dart`	Facebook/Google/Apple login options aur Sign in with password.
`signup_screen.dart`	Naya account: email aur password.
`login_screen.dart`	Existing user login.
`fill_profile_screen.dart`	Naam, nickname, DOB, phone, gender, photo.
`create_pin_screen.dart`	4-digit PIN set karna.
`fingerprint_screen.dart`	Biometric set karna.
`forgot_password_screen.dart`	Email ya SMS se reset method chunna.
`otp_screen.dart`	Code verify karna.

Screen	Kaam
`create_new_password_screen.dart`	Naya password set karna.

ViewModel: login, signup, OTP, loading aur error state. Repository: Firebase Auth ke calls.

`home/`

Screen	Kaam
`main_shell_screen.dart`	Bottom navigation bar wala container (Home, Wishlist, Cart, Orders, Profile). Baaki screens iske andar load hoti hain.
`home_screen.dart`	Greeting, search bar, offers banner, categories, most popular.
`special_offers_screen.dart`	Saare discount offers ki list.
`most_popular_screen.dart`	Popular products ka grid.

`notifications/`

Screen	Kaam
`notification_screen.dart`	Today/Yesterday/Older ke hisaab se in-app notifications.

`wishlist/`

Screen	Kaam
`wishlist_screen.dart`	Pasand kiye hue products ka grid, category chips ke saath.

`search/`

Screen	Kaam
`search_screen.dart`	Search box, recent searches, results.
`search_empty_screen.dart`	"Result nahi mila" wali state.
`sort_filter_sheet.dart`	Bottom sheet: category, price range, rating, sort, Reset/Apply.

`product/`

Screen	Kaam
`category_list_screen.dart`	Kisi bhi category (Sofa, Chair, Table, Kitchen, Lamp, Cupboard, Vase) ke products. Ek hi screen sab categories ke liye, category parameter se badalti hai.
`product_detail_screen.dart`	Bada image, description, colors, quantity, Add to Cart.
`reviews_screen.dart`	Product ke reviews aur rating filter.

`cart/`

Screen	Kaam
`cart_screen.dart`	Cart items, quantity badlo, total, Checkout button.
`remove_from_cart_sheet.dart`	"Cart se hatana hai?" confirm sheet.

`checkout/`

Screen	Kaam
`checkout_screen.dart`	Poora summary: address, items, shipping, promo, total.

Screen	Kaam
`shipping_address_screen.dart`	Saved addresses mein se chunna.
`choose_shipping_screen.dart`	Economy, Regular, Cargo, Express.
`add_promo_screen.dart`	Discount code lagana.
`payment_methods_screen.dart`	Wallet, PayPal, Google Pay, Apple Pay, card.
`enter_pin_screen.dart`	Payment ke liye PIN confirm.
`order_success_sheet.dart`	Order ho gaya wali sheet (View Order, E-Receipt).

`orders/`

Screen	Kaam
`my_orders_screen.dart`	Active aur Completed tabs.
`track_order_screen.dart`	Order ka status timeline.
`leave_review_sheet.dart`	Stars aur comment dena.

`wallet/`

Screen	Kaam
`my_wallet_screen.dart`	Balance, card aur recent transactions.
`top_up_screen.dart`	Wallet mein paise daalna.
`transaction_history_screen.dart`	Poori transaction list.
`e_receipt_screen.dart`	Barcode wali receipt.

`profile/`

Screen	Kaam
`profile_screen.dart`	Photo, naam aur settings ka menu.
`edit_profile_screen.dart`	Profile details badalna.

Screen	Kaam
`address_list_screen.dart`	Saved addresses.
`add_address_screen.dart`	Map ke saath naya address.
`notification_settings_screen.dart`	Notification on/off toggles.
`payment_settings_screen.dart`	Linked payment accounts.
`add_card_screen.dart`	Naya card add karna.
`security_screen.dart`	Biometric toggles, PIN aur password badalna.
`language_screen.dart`	Language chunna.
`privacy_policy_screen.dart`	Privacy policy ka text.
`invite_friends_screen.dart`	Contacts ko invite karna.

`help/`

Screen	Kaam
`help_center_screen.dart`	FAQ aur Contact Us tabs.
`customer_service_chat_screen.dart`	Support ke saath chat.

Naya Feature Ya Screen Kaise Add Karein

1. `features/` mein naya folder banao (jaise `features/coupons/`).
2. Uske andar `views/`, `viewmodels/`, `models/`, `repositories/` banao.
3. Repository mein pehle abstract class likho, phir Firebase wali implementation.
4. ViewModel mein state class aur Notifier likho, repository ko `ref.read` se lo.
5. Screen banao (`ConsumerWidget`) aur `ref.watch(...)` se state dikhao.
6. `core/router/app_router.dart` mein route add karo.

Galti Se Bachne Ke Rules

- Screen (View) mein Firebase ka code mat likho, wo Repository ka kaam hai.

- Screen mein logic mat likho, wo ViewModel ka kaam hai.
- Colors aur font screen mein hardcode mat karo, `AppColors` aur theme use karo.
- Jo widget do se zyada features mein chale use `shared/widgets/` mein le jao.
- `firebase_options.dart`, `google-services.json` aur keystore ko public repo mein push karne se pehle check karo.